

Evaluation of Disease Severity Using CSF Biomarkers in Patients With Probable Cerebral Amyloid Angiopathy

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Abstract

Background and Objectives

Cerebral amyloid angiopathy (CAA) is currently diagnosed using MRI-based Boston criteria. However, imaging markers represent downstream consequences of vascular injury and may not fully reflect disease severity. CSF biomarkers, particularly β -amyloid 1–42 ($A\beta_{42}$) and β -amyloid 1–40 ($A\beta_{40}$), may provide a more direct measure of vascular amyloid burden. We investigated the association between CSF biomarkers and hemorrhagic and nonhemorrhagic MRI markers of disease severity in probable CAA.

Methods

We conducted a retrospective multicenter cohort study including consecutive patients diagnosed with probable CAA according to Boston criteria v2.0 at 2 tertiary centers (2014–2023) who underwent lumbar puncture (LP) as part of clinical evaluation. CSF $A\beta_{42}$, $A\beta_{40}$, total tau, and phosphorylated-tau 181 (p-tau181) were measured using standardized immunoassays. MRI markers included lobar cerebral microbleeds (CMBs), cortical superficial siderosis (cSS), white matter hyperintensity burden, Fazekas score, and enlarged perivascular spaces in the centrum semiovale. Multivariable linear regression models were adjusted for age, sex, MRI sequence type, and delay between onset and LP. False discovery rate correction was applied for multiple testing. We also used principal component analysis to explore associations between MRI and CSF biomarkers.

Results

A total of 102 patients were included (mean age at LP 72.0 ± 7.5 years; 65% male). Initial presentations were cognitive impairment in 53%, intracerebral hemorrhage in 34%, and transient focal neurologic episodes in 13%. Lower CSF $A\beta_{40}$ and $A\beta_{42}$ levels were significantly associated with greater cSS burden (standardized β -0.49 [95% CI -0.8 to -0.17], $p = 0.002$ and -0.42 [-0.73 to -0.11], $p = 0.007$, respectively) and higher Fazekas score (β -0.18 [-0.33 to -0.03], $p = 0.02$ and -0.23 [-0.38 to -0.08], $p = 0.002$) but not with CMB count. No significant association was found between tau species and MRI markers. Associations remained significant after stratification by CSF $A\beta_{42}/A\beta_{40}$ -defined Alzheimer-like profile.

Discussion

In probable CAA, lower CSF $A\beta_{40}$ and $A\beta_{42}$ levels are associated with established MRI markers of disease severity, independently of clinical presentation and concomitant Alzheimer-like profile. These findings suggest that CSF $A\beta$ levels may reflect vascular amyloid burden rather than downstream neurodegeneration. Prospective longitudinal studies are needed to determine their prognostic value.

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Supplementary Material

Glossary

A β 40 = amyloid β 1–40; **A β 42** = amyloid β 1–42; **AD** = Alzheimer disease; **ARIA** = amyloid-related imaging abnormalities; **CAA** = cerebral amyloid angiopathy; **CMB** = cerebral microbleed; **CSO** = centrum semiovale; **cSS** = cortical superficial siderosis; **ePVS** = enlarged perivascular spaces; **FLAIR** = fluid-attenuated inversion recovery; **GRE** = gradient echo; **ICD-10** = International Classification of Diseases, 10th Revision; **ICH** = intracerebral hemorrhage; **LP** = lumbar puncture; **PCA** = principal component analysis; **p-tau181** = phosphorylated tau 181; **STRIVE** = Standards for Reporting Vascular changes on Neuroimaging; **SWAN** = susceptibility-weighted angiography; **SWI** = susceptibility-weighted imaging; **TFNE** = transient focal neurologic episodes; **t-tau** = total tau; **WMH** = white matter hyperintensity.

Introduction

In vivo diagnosis of sporadic cerebral amyloid angiopathy (CAA) relies on brain MRI using the Boston criteria, which identify characteristic hemorrhagic markers such as lobar intracerebral hemorrhage (ICH), cerebral microbleeds (CMB), and cortical superficial siderosis (cSS) on susceptibility-sensitive sequences. The most recent Boston criteria v2.0 (2022) expanded this framework by incorporating non-hemorrhagic markers—including enlarged perivascular spaces in the centrum semiovale (CSO-ePVS) and a multi-spot pattern of cortical-subcortical white matter hyperintensities (WMH)—which improve diagnostic sensitivity and specificity.¹

Although these neuroimaging markers have strengthened diagnostic accuracy, they largely represent downstream consequences of vascular amyloid injury. As such, they may fail to capture early or presymptomatic disease stages and provide only an indirect reflection of underlying pathophysiology. CSF biomarkers offer a complementary strategy by measuring molecular signatures that are more directly linked to vascular amyloid deposition and clearance dysfunction.² The CSF levels of β -amyloid 1–42 (A β 42), β -amyloid 1–40 (A β 40), total tau (t-tau) protein and its phosphorylated form at threonine 181 (p-tau181) can be assessed in clinical care. The association between vascular amyloid deposition and reduced CSF A β levels in CAA, particularly A β 40, was highlighted in several studies.^{3,4} Nonetheless, they are not included in the most recent version of Boston criteria because of insufficient data. The sensitivity of these criteria differs according to the clinical presentation, the diagnostic accuracy being mainly driven by hemorrhagic manifestations.⁵ In contrast, CSF biomarkers showed alterations regardless of clinical phenotype.² CSF biomarkers may contribute to differentiating CAA from Alzheimer disease (AD), although they do not allow a direct distinction between the 2 conditions.⁴ This distinction is clinically important given the expanding use of anti-amyloid monoclonal antibodies for AD and the high risk of amyloid-related imaging abnormalities (ARIA) in individuals with underlying CAA.⁶

Our study examined correlations between CSF biomarkers and established MRI markers, both hemorrhagic and non-hemorrhagic in patients with probable CAA to better clarify their potential contribution to diagnosis. In addition, we

aimed to investigate the associations between various clinical manifestations of CAA and CSF.

Methods

Patient Inclusions

We conducted a retrospective multicenter study across 2 tertiary referral memory and stroke centers in France (Rouen and Tours University Hospitals) between 2014 and 2023. Consecutive patients fulfilling the Boston criteria v2.0 for probable CAA were identified through systematic screening of hospital databases using the ICD-10 diagnostic code for CAA. Patients were included if they had both CSF biomarker analysis obtained by lumbar puncture (LP) and available brain MRI data, regardless of their initial clinical presentation. Patients with only possible CAA or with CAA-related inflammation were excluded because of diagnostic uncertainty and their distinct CSF profiles, respectively.⁷

In our centers, CSF analysis is proposed during the etiologic work-up of suspected CAA regardless of their clinical phenotype, including patients presenting with ICH, transient focal neurologic episodes, or cognitive symptoms, both to increase diagnostic confidence and exclude alternative diagnoses. In patients presenting with cognitive symptoms, LP is routinely performed according to French national recommendations for the diagnostic evaluation of suspected AD.⁸ LP was also proposed to patients presenting with ICH, transient focal neurologic episodes (TFNE), or other CAA-related manifestations, when clinically appropriate. To avoid transient CSF biomarker changes, LP was performed at least 3 months after any cerebrovascular event, head trauma, or seizure, given the known transient increase in CSF tau levels in these conditions.⁹ Clinical presentation was categorized into 3 groups: ICH, TFNE (defined as recurrent and stereotypical episodes with positive seizure-like or negative transient ischemic attacks-like symptoms), and progressive cognitive decline.

CSF Analyses

To minimize intersite variability in CSF measurements, both centers collected CSF in a standardized 10-mL polypropylene tube (catalog number 62.610.201; Sarstedt, Nümbrecht, Germany). Preanalytical procedures, from LP to delivery to the laboratory for processing, were harmonized across the 2

sites. Following centrifugation for 10 minutes at +4°C, samples were aliquoted into 2-mL polypropylene cryotubes and frozen at −80°C within 4 hours. CSF biomarkers (Aβ42, t-tau, and p-tau181) were measured at both centers either with ELISA-based INNOTEST kits (Fujirebio Europe, Ghent, Belgium) in a semiautomated manner or using fully automated immunoassays on the Lumipulse G600II platform with commercially available Fujirebio kits. CSF Aβ40 was measured exclusively on the Lumipulse G600II platform Aβ40, either concurrently with Aβ42, Tau, and p-tau181 or later, using a previously unfrozen CSF aliquot. For ELISA assays, all measurements were performed in duplicate, and a coefficient of variation below 15% was considered acceptable; samples exceeding this threshold were reanalyzed in duplicate. Both centers followed identical analytical protocols and quality control procedures.

At both centers, an AD CSF profile was defined as the combination of low CSF Aβ42 levels (i.e., <635) with either increased t-tau levels (i.e., >466) and/or increased p-tau181 (i.e., >55).

Before statistical analyses, all biomarker measurements were log-transformed and scaled after log-transformation. Aβ42 measurements were log-transformed and scaled separately for ELISA and Lumipulse measurements.

MRI Analyses

MRI examinations were performed in the 2 participating tertiary centers using 1.5T or 3T scanners as part of routine clinical care. Imaging protocols were not fully standardized owing to the retrospective multicenter design and the extended inclusion period, during which clinical imaging practices evolved over time, but consistently included at least fluid-attenuated inversion recovery (FLAIR) imaging and a blood-sensitive sequence in all included patients. One neuroradiologist (Q.B.) analyzed all anonymized MRI images using a standardized extraction form while blinded to clinical and CSF findings. A second neuroradiologist (D.T.) assessed a subgroup of 30 cases to evaluate interrater agreement (eTable 1).

Infarcts, WMH, and CMB were evaluated in accordance with the Standards for Reporting Vascular changes on Neuroimaging (STRIVE) criteria.¹⁰ Acute DWI-positive lesions were defined as a hyperintense signal on diffusion-weighted imaging sequences with corresponding restricted diffusion on apparent diffusion coefficient maps. The presence of FLAIR “multispot pattern” was noted. Chronic WMH were graded in the periventricular and deep white matter according to the Fazekas scale (total score range 0–3),¹¹ and a total WMH burden in milliliters was estimated through semiautomatic lesion segmentation using the level tracing tool from 3D Slicer software v5.2.2 based on FLAIR sequence images.¹²

To assess hemorrhagic features, we prioritized susceptibility-weighted imaging (SWI) or susceptibility-weighted angiography

(SWAN) but used T2* or gradient echo (GRE) sequences when the latter were the only blood-sensitive sequence available. CMBs were defined as small (2–10 mm) round foci of hypointense signals on blood-sensitive sequences and were allocated to either deep or lobar locations in accordance with the Microbleed Anatomical Rating Scale.¹³ Following the Boston Criteria v2.0, only strictly lobar microbleeds were considered for the diagnosis of probable CAA. Lobar CMBs were defined as microbleeds located in cortical or immediately subcortical regions of the cerebral lobes. According to Boston Criteria, the presence of any CMB in deep brain territories precluded the diagnosis of probable CAA, and consequently, the patient was excluded from the study. Parenchymal hemorrhagic lesions larger than 10 mm were considered as macrohemorrhages. Acute subarachnoid hemorrhage (SAH) was defined as a hyperintense signal on FLAIR sequences within 1 or multiple brain sulci. cSS was defined as homogenous hypointense curvilinear signal losses on blood-sensitive sequences that outline the outer surface of the cerebral cortex, within the adjacent subarachnoid space or both.¹⁴ cSS was rated according to the cSS multifocality scale.¹⁵ Hemorrhagic lesions were counted based on a visual analysis. Cerebral atrophy was evaluated visually using the simplified global cerebral atrophy, which is based on the widening of the cerebral sulci.¹⁶ ePVS were visually rated in the basal ganglia and the CSO using a validated 4-point semiquantitative scale.¹⁷ In line with the Boston Criteria v2.0, a burden of ≥20 ePVS in the CSO was considered as indicative of a high ePVS score. In accordance with STRIVE recommendations, ePVS were rated preferentially on T2-weighted images when available, and on 3D T1-weighted images otherwise, acknowledging reduced sensitivity with the latter approach. However, ePVS assessment was not systematically feasible across the entire cohort owing to the lack of appropriate sequences in a subset of patients. The CAA score was also rated as a marker of the disease severity.¹⁸

Statistical Analyses

Descriptive statistics were expressed as mean with SD and median. We used the Wilcoxon Mann-Whitney test or Kruskal-Wallis for pairwise comparisons for quantitative data and a Fisher exact test for qualitative data. CSF biomarker levels were compared between high and low ePVS counts by Student *t* tests. The distribution of CSF biomarker levels was represented by raincloud plots according to the initial clinical presentation (dichotomized into ICH or TFNE for hemorrhagic manifestations and cognitive decline).

Pearson correlation coefficients were computed between each MRI marker or score and scaled log-transformed CSF biomarkers and are shown in a correlation matrix. Multiple testing among correlation coefficients was accounted for by false discovery rate correction (Benjamini-Hochberg). All nominally significant correlation values ($p < 0.05$) were colored according to the strength and direction of the association. A threshold of 5% FDR was applied to highlight significant correlation values after multiple testing correction in bold font.

Linear regression coefficients of MRI markers or score on scaled log-transformed CSF biomarkers were adjusted for delay from disease onset to LP, magnetic blood-susceptibility sequence used for MRI markers (T2* vs SWI/SWAN), age at onset, and sex. Corresponding *p*-values and 95% CIs were computed along with FDR values to correct for multiple testing. In both correlation and adjusted linear regression analyses, multiple testing correction was accounted for all investigated biomarker-MRI marker pairs.

Finally, a principal component analysis (PCA) of MRI markers and CSF biomarkers was computed to visualize the overall structure of the data set and identify underlying patterns of covariance, notably with ePVS which has been underlined by linear regression. This data-driven approach transforms correlated variables into a reduced set of orthogonal components, each representing a distinct combination of biomarkers that capture shared variance across measures. First components capture the strongest signals of variation while following components each capture finer details. In our analysis, each of these principal components is defined by a linear combination of biomarkers and can therefore be interpreted in terms of the biomarkers that contribute the most to it. Therefore, patients were projected into the first 2 axes and colored according to ePVS scores. This first plan can be interpreted according to the contribution of each MRI and CSF marker to the construction of these axes (see eMethods for further details).

Standard Protocol Approvals, Registrations, and Patient Consents

Following the current French legislation regarding retrospective study, we excluded any patient mentioning his opposition for the use of his CSF sample for research purposes. Ethics board approval was obtained for data collection and publication from the institutional review board of Rouen University Hospital for retrospective noninterventional research (E2025-48).

Data Availability

All data generated or analyzed during this study are included in this published article and its supplementary material files.

Results

Probable CAA Population and CSF Biomarker Profile

We included 102 patients with probable CAA (65% male) (Table 1, Figure 1) recruited in the centers of Rouen University Hospital (*n* = 64) or the Tours Hospital (*n* = 38). The mean age at presentation was 69.9 ± 7.5 years, and the mean age at LP was 72.0 ± 7.1 years. The most common initial presentation was cognitive decline (52.9%, *n* = 54), followed by ICH (34.3%, *n* = 35) and TFNE (12.7%, *n* = 13). Among patients with TFNE, 6 showed acute SAH and 7 cSS. The Mini-Mental Status Evaluation at LP was available in 80/102 patients with an average score of 22.7 ± 5.6 . APOE genotype

was available for a minority of patients (39/102, 38.2%). Among those 39 patients, 8 (20.5%) carried at least 1 APOE ε2 allele and 18 (46.1%) at least 1 APOE ε4 allele.

Median CSF biomarker concentrations were $A\beta_{42} = 462.0 \pm 231.1$ pg/mL, $A\beta_{40} = 9,048.6 \pm 2,965.1$ pg/mL, total tau = 518 ± 363.7 pg/mL, and p-tau181 = 74.3 ± 44.0 pg/mL. Demographic and biomarker distributions across clinical presentation groups (cognitive onset vs ICH/TFNE) are summarized in Table 1 and eFigure 1. Overall, 89 patients showed low CSF Aβ42 levels and 55 patients (53.9% of the total cohort) also showed increased t-tau and/or p-tau181 levels. Among them, 31 patients presented with cognitive decline, whereas 15 patients presented with ICH and 9 with TFNE.

Age at onset did not differ between groups, although the interval from onset to LP was significantly longer among patients with cognitive presentations (2.9 ± 3.2 vs 1.7 ± 2.5 years, *p* = 0.01). Patients presenting with cognitive decline exhibited significantly higher CSF t-tau and p-tau181 levels (t-tau: 588.6 ± 410.1 vs 441.3 ± 289.1 pg/mL, *p* = 0.01; p-tau181: 85.2 ± 53.7 vs 62.0 ± 24.8 pg/mL, *p* = 0.03), whereas Aβ42 and Aβ40 concentrations did not differ by clinical phenotype.

Correlation Between CSF Biomarkers and Neuroimaging Markers

The median interval between LP and MRI was 3.4 ± 14.9 months (Table 1). 1.5T MRI were rated in 60 patients and 3 Tesla MRI in 42 patients. All patients but 5 showed lobar CMBs (median count: 17 ± 180.5), cSS was present in 60 patients (58.8%), including 36 with disseminated involvement. CSO-ePVS were rated in only 77 (75.5%) patients given the availability of T2-weighted sequence in 37 patients and 3D-T1 in 40 patients. Nine patients presented with <10 ePVS, 11 with 11–20 ePVS, 31 with 21–40 ePVS, and 26 with >40 ePVS.

Patients with initial hemorrhagic presentations had significantly higher rates of disseminated cSS (45.8% vs 27.7%, *p* = 0.05). By contrast, cognitive-onset patients showed higher microbleed counts (median ± SD: 42 ± 195.2 vs 9 ± 158.0 , *p* = 0.004) and greater WMH burden (37.6 ± 29.7 vs 25.9 ± 22.7 , *p* = 0.02).

A correlation matrix using Pearson coefficients with false discovery rate correction (Figure 2) demonstrated significant negative associations between CSF Aβ40 and Aβ42 levels and cSS burden (Aβ40: *r* = −0.33, *p* < 0.001; Aβ42: *r* = −0.26, *p* = 0.01). No correlations were observed between Aβ biomarkers and total CMB counts, except for the Aβ42/Aβ40 ratio (*r* = −0.23, *p* < 0.01).

Both Aβ40 and Aβ42 were negatively correlated with non-hemorrhagic markers, including Fazekas WMH scores (Aβ40: *r* = −0.27, *p* = 0.006; Aβ42: *r* = −0.34, *p* < 0.001). WMH volume also showed a significant negative correlation (Aβ40: *r* = −0.20, *p* = 0.04; Aβ42: *r* = −0.26, *p* = 0.008).

No significant correlations were found between tau or p-tau181 levels and any MRI markers.

Table 1 Comparison Between Patients With CAA According to Their Initial Clinical Presentation

	Total included patients (n = 102)	CAA presenting with cognitive impairment (n = 54)	CAA presenting with ICH or TFNE (n = 48)	p Value ^a
Sex, male, n (%)	65 (63.7)	38 (70.3)	27 (56.2)	0.15
Age at onset (y), mean ± SD	69.9 ± 7.5	69.7 ± 7.2	70.0 ± 7.9	0.46
Age at LP (y), mean ± SD	72.0 ± 7.1	72.3 ± 6.3	71.6 ± 7.9	0.60
Time from onset to LP (y), mean ± SD	2.39 ± 2.9	2.9 ± 3.2	1.7 ± 2.5	0.01
High blood pressure, n (%)	70 (68.6)	32 (59.2)	38 (79.1)	0.03
Aβ42 (pg/mL), mean ± SD	462.0 ± 231.1	457.0 ± 231.3	416.5 ± 234.3	0.58
Aβ40 (pg/mL), mean ± SD	9,048.6 ± 2,965.1	8,968.1 ± 2,688.9	9,139.5 ± 3,277.4	0.88
Tau protein (pg/mL), mean ± SD	518 ± 363.7	588.6 ± 410.1	441.3 ± 289.1	0.01
p-tau181 protein (pg/mL), mean ± SD	74.3 ± 44.0	85.2 ± 53.7	62.0 ± 24.8	0.03
Blood-susceptibility sequences used (T2*/SWI or SWAN)	32/70	19/35	12/35	0.38
Total number of lobar microbleeds, median ± SD	17 ± 180.5	42 ± 195.2	9 ± 158.0	0.004
Presence of cSS, n (%)	60 (58.8)	27 (50.0)	33 (68.7)	0.07
Disseminated cSS, n (%)	36 (35.3)	15 (27.7)	22 (45.8)	0.05
Presence of multispot pattern of WMH, n (%)	58 (56.8)	36 (66.6)	22 (45.8)	0.07
WMH burden (mL), mean ± SD	32.1 ± 26.7	37.6 ± 29.7	25.9 ± 22.7	0.02
Presence of DWI-positive lesions, n (%)	17 (16.5)	6 (11.1)	11 (22.9)	0.12
ePVS in semiovale center, n (%)				0.20
Available for rating	77 (75.5)	41 (75.9)	36	
<10 ePVS	9	4	5	
11–20 ePVS	11	7	4	
21–40 ePVS	31	20	11	
>40 ePVS	26	10	16	

Abbreviations: Aβ40 = amyloid β1–40; Aβ42 = amyloid β1–42; CAA = cerebral amyloid angiopathy; cSS = cerebral superficial siderosis; DWI = diffusion-weighted imaging; ePVS = enlarged perivascular spaces; ICH = intracerebral hemorrhage; LP = lumbar puncture; p-tau = phosphorylated tau; SWAN = susceptibility-weighted angiography; SWI = susceptibility-weighted imaging; tau = tau protein; TFNE = transient focal neurologic episodes; WMH = white matter hyperintensity.

^a p Value of statistical test comparing CAA with initial cognitive impairment vs CAA with initial ICH or TFNE using Wilcoxon Mann-Whitney test or Kruskal-Wallis for continuous variables and the Fisher exact test for categorical variables.

These findings remained robust in linear regression models adjusting for age at onset, sex, time from onset to LP, and MRI sequence type (T2* vs SWI/SWAN) (Figure 3).

We then performed stratified supplemental analyses using validated Aβ42/Aβ40 ratio cutoffs (i.e., 0.055). Fifty-two patients met AD CSF criteria. Among them, 30 patients presented with cognitive decline but 15 with ICH and 7 with TFNE. In the subgroup with normal ratio, Aβ42 remained significantly associated with cSS and Fazekas score (eFigure 2).

Principal Component Analysis of MRI and CSF Biomarkers

To further investigate associations involving CSO-PVS, a PCA was performed (eFigure 3 and eTable 2). The first 2

principal components accounted for 40.5% of total variance. Interpretation of variable contributions to PCs 1 revealed a strong inverse correlation between MRI severity markers (Fazekas WMH score and WMH burden) with Aβ42 levels. Axis 2 revealed an independent strong inverse relationship between CSF Aβ40 and combined MRI severity markers—including Fazekas WMH score, cSS, and ePVS burden. Accordingly, patients positioned on the left-hand side of the PCA plot exhibited higher CAA severity, more extensive cSS, greater WMH and ePVS burden, and lower CSF Aβ concentrations. This multivariate pattern reinforces the univariable findings that reduced CSF Aβ40 and Aβ42 levels are strongly associated with greater overall CAA-related structural injury. We also observed “clustering” of patients with low CSO-PVS burden located in the lower half of the PCA plot, in contrast to high CSO-PVS burden clustered in the upper half.

Figure 1 Flowchart

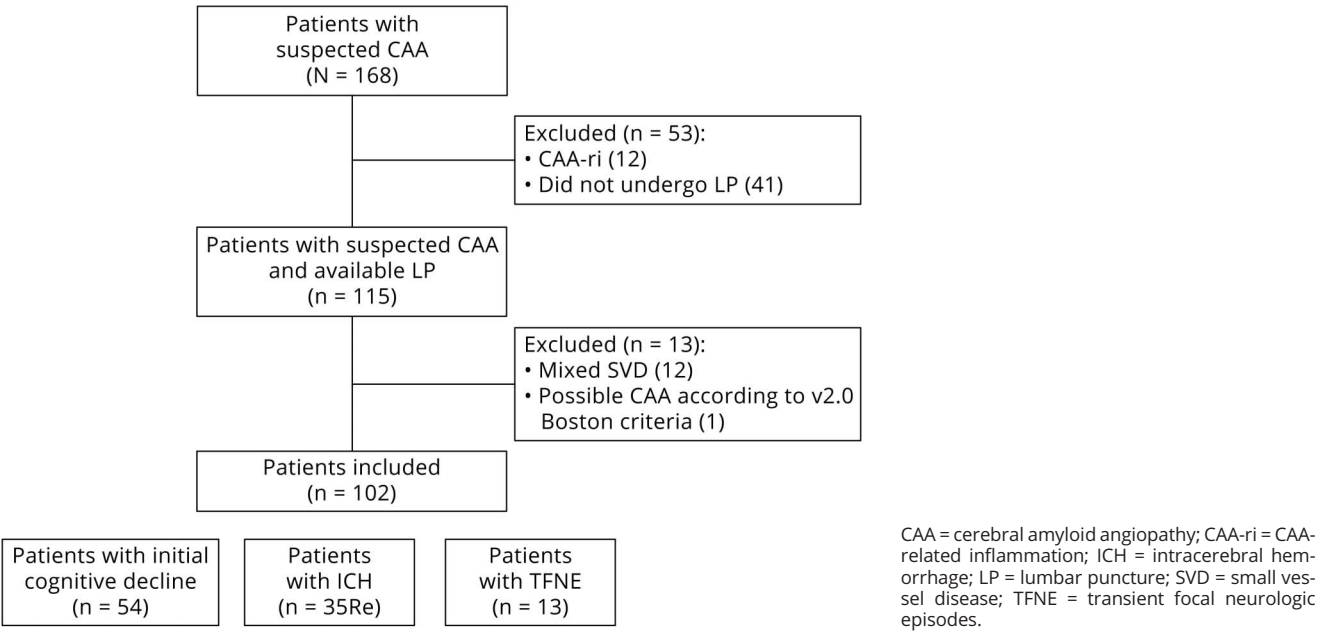
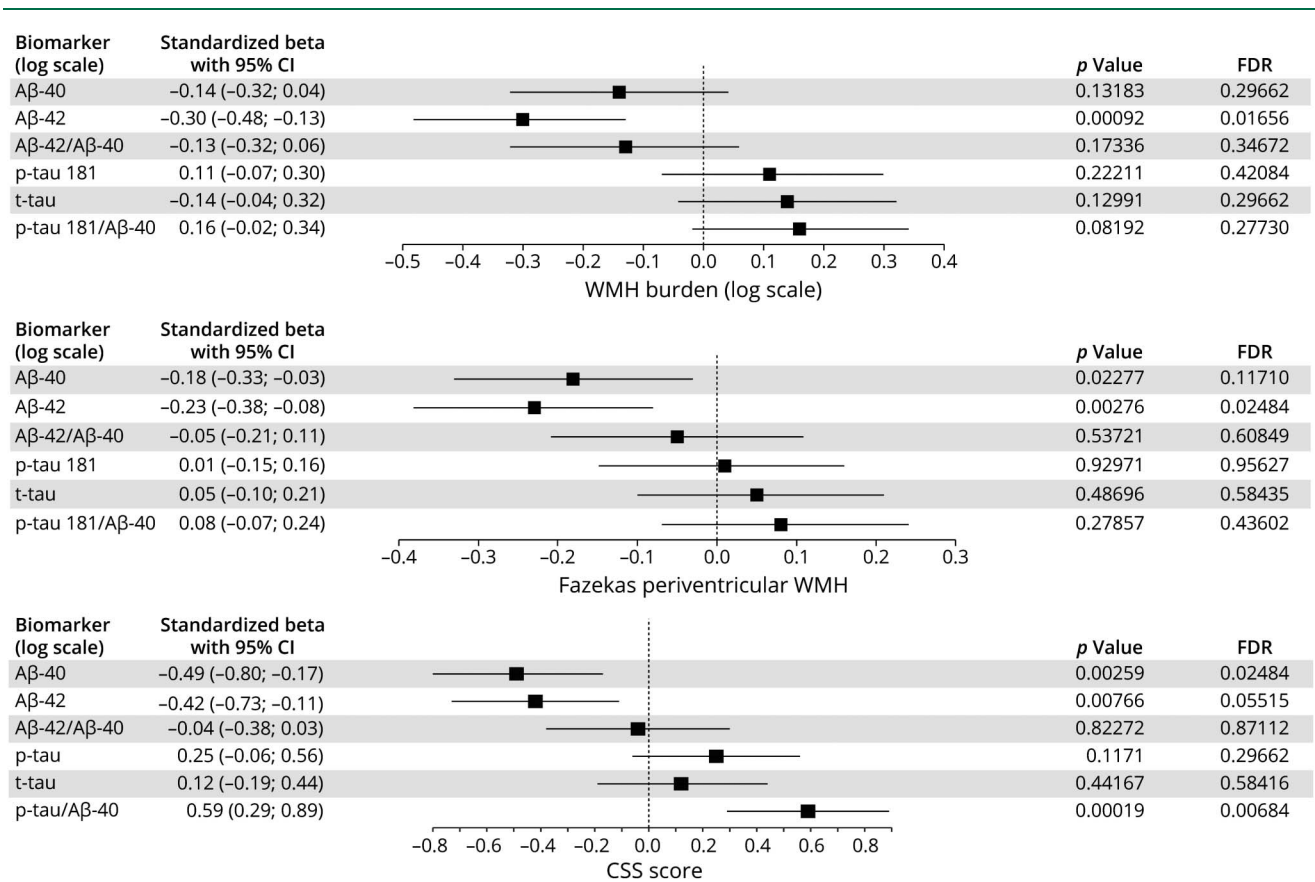


Figure 2 Correlations Between MRI and CSF Biomarkers According to Pearson Coefficients

All patients with CAA							
MRI	WMH burden	-0.204 <i>p</i> = 0.0443	-0.26 <i>p</i> = 0.0082	-0.081 <i>p</i> = 0.4277	0.064 <i>p</i> = 0.5228	0.134 <i>p</i> = 0.1871	0.081 <i>p</i> = 0.4196
	Total superficial lobar CMBs (log scale)	0.079 <i>p</i> = 0.4443	-0.191 <i>p</i> = 0.0560	-0.238 <i>p</i> = 0.0188	0.112 <i>p</i> = 0.2633	0.078 <i>p</i> = 0.4485	0.078 <i>p</i> = 0.4408
	Simplified global atrophy scale (sGCA)	-0.165 <i>p</i> = 0.1044	-0.107 <i>p</i> = 0.2847	-0.001 <i>p</i> = 0.9957	0.061 <i>p</i> = 0.5453	0.155 <i>p</i> = 0.1278	0.142 <i>p</i> = 0.1542
	Lobar lacunes	-0.123 <i>p</i> = 0.2292	-0.092 <i>p</i> = 0.3603	0.043 <i>p</i> = 0.6719	-0.028 <i>p</i> = 0.7776	0.052 <i>p</i> = 0.6124	0.008 <i>p</i> = 0.9351
	Flair multispot WMH	0.034 <i>p</i> = 0.7431	-0.185 <i>p</i> = 0.0621	-0.213 <i>p</i> = 0.0348	0.088 <i>p</i> = 0.3815	0.094 <i>p</i> = 0.3566	0.159 <i>p</i> = 0.1107
	Fazekas periventricular WMH	-0.273 <i>p</i> = 0.0065	-0.34 <i>p</i> = 0.0005	-0.077 <i>p</i> = 0.4483	0.018 <i>p</i> = 0.8574	0.15 <i>p</i> = 0.1397	0.081 <i>p</i> = 0.4209
	ePVS	-0.128 <i>p</i> = 0.3046	0.036 <i>p</i> = 0.7645	0.176 <i>p</i> = 0.1574	-0.213 <i>p</i> = 0.0765	-0.116 <i>p</i> = 0.3532	-0.175 <i>p</i> = 0.1476
	DWI count	0.018 <i>p</i> = 0.8666	0.086 <i>p</i> = 0.4225	0.091 <i>p</i> = 0.4023	-0.014 <i>p</i> = 0.8996	-0.019 <i>p</i> = 0.8613	-0.022 <i>p</i> = 0.8366
	CSS score	-0.332 <i>p</i> = 0.0009	-0.255 <i>p</i> = 0.0101	0.017 <i>p</i> = 0.8657	0.134 <i>p</i> = 0.1814	0.357 <i>p</i> = 0.0003	0.05 <i>p</i> = 0.6205
		Aβ-40	Aβ-42	Aβ-42/Aβ-40	p-tau 181	p-tau 181/Aβ-40	t-tau
CSF							

Each colored box indicates a *p* value <0.05 and each bold box indicates a false discovery rate value <0.05 after multiple test correction. Aβ40 = amyloid β1–40; Aβ42 = amyloid β1–42; CAA = cerebral amyloid angiopathy; CMB = cerebral microbleed; CSS = cortical superficial siderosis; DWI = diffusion-weighted imaging; ePVS = enlarged perivascular spaces in centrum semiovale; p-tau181 = phosphorylated tau 181; sGCA = Global Cerebral Atrophy Scale; t-tau = total tau; WMH = white matter hyperintensity.

Figure 3 Adjusted Standardized Beta IRM as Function of CSF Biomarkers



Adjusted for sex, age at onset, time from onset to LP and type of MRI sequence (T2* vs SWI/SWAN weighted-sequences). A β 40 = amyloid β 1-40; A β 42 = amyloid β 1-42; CSS = cortical superficial siderosis; FDR = false discovery rate; IRM = immunoreactive material; LP = lumbar puncture; p-tau181 = phosphorylated tau 181; SWAN = susceptibility-weighted angiography; SWI = susceptibility-weighted imaging; t-tau = total tau; WMH = white matter hyperintensity.

Discussion

This study suggests that CSF A β 40 and A β 42 concentrations are robust correlates of radiologic disease severity in probable CAA. Across a deeply phenotyped cohort, lower CSF A β species consistently tracked with both hemorrhagic (cSS) and nonhemorrhagic (WMH, Fazekas) markers, supporting the concept that CSF A β depletion reflects vascular amyloid burden more globally. These results build on emerging evidence that CSF biomarkers can provide meaningful biologic staging in CAA and may ultimately contribute to improved biologic characterization of disease severity.

We observed strong negative correlations between CSF A β 40 and A β 42 levels and cSS severity. Beyond its diagnostic value, cSS represents an active and progressive marker of leptomeningeal vessel fragility and amyloid-related vasculopathy, particularly when disseminated.^{19,20} Our findings align with this interpretation: reduced CSF A β likely reflects sequestration of soluble peptides within the vessel wall, consistent with impaired clearance and increased deposition. Similar associations were seen with nonhemorrhagic MRI markers introduced in the Boston v2.0 criteria—including WMH burden and Fazekas scores—largely believed to reflect CAA-related mechanisms

such as chronic hypoperfusion, impaired perivascular clearance, and blood-brain barrier dysfunction secondary to vascular amyloid deposition, rather than primary neurodegenerative processes as seen in AD.^{10,21} Together, these results support the view that CSF A β concentrations index the cumulative small-vessel injury characteristic of CAA rather than the downstream hemorrhagic sequelae alone.

Although previous studies reported similar trends, most were limited by small sample sizes or incomplete biomarker panels.^{22,23} For example, 1 study found reduced A β 42 associated with cSS but lacked A β 40 measurements, limiting pathophysiologic interpretation.²³ Our larger cohort with standardized imaging strengthens the evidence that CSF A β 40 or A β 42 reductions mirror overall disease severity. Notably, we found an inverse correlation between total number of lobar CMBs and CSF A β 42/A β 40 ratio but without correlation with each specific A β levels. One possible explanation for the inverse association between CSF A β 42/A β 40 ratio and lobar CMB burden is that a low ratio identifies a subgroup of CAA patients enriched for concomitant AD-type parenchymal amyloid pathology. In such patients, vascular amyloid-related hemorrhagic injury and additional A β 42 sequestration into plaques may coexist, leading to both a greater lobar CMBs

burden and a more AD-like CSF profile.²⁴ An alternative, nonexclusive explanation is that this pattern reflects an *APOE* $\epsilon 4$ -associated capillary CAA phenotype, lying on the biologic continuum between sporadic CAA and AD as *APOE* $\epsilon 4$ remained the main genetic risk factor for AD.²⁵ In the end CMBs are focal, heterogeneous lesions that may arise from mixed etiologies, can remain clinically silent, and do not always reflect the degree of amyloid infiltration.²⁶⁻²⁹ These findings are in line with a recent prospective Italian study comparing CAA patients with and without prior symptomatic ICH, which showed significantly lower CSF A β 40 levels in hemorrhagic CAA, while other CSF biomarkers did not differ between groups.³⁰ Together, these converging results support the main hypothesis that CSF A β 40 depletion reflects increased vascular amyloid deposition and may be specifically associated with a hemorrhagic CAA phenotype.

The PCA revealed a subtler but mechanistically meaningful association between severely reduced A β 40 and high CSO-PVS burden. Patients clustering in the high-PVS/low-A β 40 region exhibited combined radiologic severity across CAA domains. These findings align with the hypothesis that impaired perivascular clearance simultaneously promotes A β 40 retention in the vessel wall, reduces soluble CSF A β , and drives enlargement of perivascular spaces. Recent observational and mechanistic data increasingly support this framework, and our results add evidence that CSF A β 40 may serve as a biologic signature of clearance dysfunction in sporadic CAA.²¹

Despite the heterogeneity of clinical presentation, CSF biomarker profiles remained consistent, with similarly altered A β levels irrespective of whether the presentation was primarily cognitive or hemorrhagic. This contrasts with radiologic markers, which can differ according to initial or predominant CAA presentation. In our cohort, patients with CAA with an initial cognitive (i.e., nonhemorrhagic) presentation had fewer putative hemorrhagic CAA markers compared with those presenting with ICH, despite all patients fulfilling radiologic criteria for probable CAA. These results suggest that CSF A β 40 and A β 42 may serve as a phenotype-independent marker of vascular amyloid load, suggesting potential utility as a complementary biologic marker even when classical hemorrhagic features required by Boston v2.0 are absent. This is particularly relevant for cognitive-predominant CAA, where diagnostic sensitivity remains suboptimal.

Unlike A β , CSF tau and p-tau181 levels varied by clinical phenotype. Patients presenting with cognitive impairment exhibited significantly higher tau markers, while tau did not correlate with MRI markers of CAA burden. This pattern likely reflects coexisting AD pathology rather than CAA-related neurodegeneration, consistent with the distinct biochemical profiles of AD and CAA.² Recent imaging-CSF studies similarly demonstrated distinct signatures of cerebral atrophy in the presence of an AD-like CSF profile.³¹ However, these results remain controversial as CSF biomarker profile associated with AD in patients with sporadic CAA did not

affect global cognition, specific cognitive domains, or the presence of neuropsychiatric symptoms in another study.³² Future studies should investigate further the clinical significance of AD specific markers in CAA using complementary approaches such as tau-PET imaging.

Because CSF A β 40 and A β 42 concentrations were associated with disease severity across clinical phenotypes, they may help refine biologic assessment in scenarios in which Boston v2.0 criteria underperforms, such as cognitive-only presentations. Several frameworks—including recent iatrogenic CAA criteria—have proposed incorporating CSF A β ratios into multimodal assessment strategies.^{33,34} CSF biomarkers may also reflect early, potentially preradiologic disease stages, as suggested in hereditary CAA where CSF changes precede MRI abnormalities.³⁵

These biomarkers also carry important implications for AD immunotherapy safety.⁶ ARIA risk correlates strongly with cerebrovascular amyloid burden. Recent mechanistic work suggests that antibody mobilization of vascular A β causes focal endothelial injury, leading to vasogenic edema.^{6,36-38} Individuals with extremely low CSF A β 40—reflecting high vascular A β saturation—may be biologically primed for ARIA.

Strengths of our study include its relatively large CAA cohort, harmonized CSF immunoassay pipeline, centralized imaging rating, and integration of clinical, radiologic, and biologic phenotypes. Several limitations must be acknowledged. The retrospective design and heterogeneous MRI-to-LP intervals could introduce noise, although adjusted models confirmed the robustness of correlations. In addition, selection bias cannot be excluded, as CSF analysis was not systematically performed but rather guided by clinical indications, potentially leading to an overrepresentation of patients with cognitive symptoms or diagnostic uncertainty. Moreover, the study was conducted in tertiary referral centers and required the availability of both MRI and CSF data, which may have led to an overrepresentation of more extensively investigated or complex cases. These factors may limit the generalizability of our findings to the broader CAA population. Mixed blood-sensitive MRI sequences (SWI/SWAN vs GRE) may influence detection sensitivity; however, we adjusted for sequence type and used experienced raters. CSO-ePVS scoring was performed using T2 or 3D-T1 because of sequence availability; although acceptable, this approach is less standardized. These methodologic considerations underscore the need for prospective studies with fully harmonized MRI protocols and predefined biomarker sampling windows. A further limitation relates to the incomplete availability of *APOE* genotype, which was obtained in only a subset of patients and precluded meaningful adjustment or stratified analyses. Given the established role of *APOE* $\epsilon 2$ and $\epsilon 4$ alleles in modulating CAA severity and hemorrhagic risk, residual confounding cannot be excluded and future studies incorporating systematic genetic data will be important to clarify the interplay between *APOE* status and CSF biomarkers.

In conclusion, CSF A β 40 and A β 42 concentrations closely mirror cumulative radiologic markers of CAA severity and represent a biologically grounded, phenotype-independent measure of vascular amyloid load. Incorporating CSF biomarkers into multimodal frameworks could help refine disease characterization for early-stage and cognitively predominant presentations, guide referral to specialized centers, and refine ARIA risk assessment before AD antibody therapy. At the bedside, a markedly low A β 40 profile may serve as a marker of high vascular amyloid burden and could help identify patients potentially at an increased risk of hemorrhagic complications.

Author Contributions

Q. Beaufort: drafting/revision of the manuscript for content, including medical writing for content. C. Charbonnier: analysis or interpretation of data. P. Ruellan: major role in the acquisition of data. O. Ozkul-Wermester: major role in the acquisition of data. V. Amar: major role in the acquisition of data. D. Turpin: major role in the acquisition of data. M. Quillard-Muraine: major role in the acquisition of data. E. Gerardin: analysis or interpretation of data. D. Dufour: analysis or interpretation of data. A. Charidimou: study concept or design. A.-C. Balageas: study concept or design. G. Boulouis: drafting/revision of the manuscript for content, including medical writing for content. M. Pasi: drafting/revision of the manuscript for content, including medical writing for content. D. Wallon: drafting/revision of the manuscript for content, including medical writing for content. L. Grangeon: drafting/revision of the manuscript for content, including medical writing for content.

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